

Cell, Aspect, Self: The Three-Level Structural Framework as Architectural Primitive in the Coordination Knowledge Substrate Pattern

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in *The Instinct/Reasoning Separation Outside the Model* (Li, April 2026), the second paper in the CKS theory series following *Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems* (Li, April 2026). It does not introduce new axioms. Its sole contribution is to formalize the three-level structural framework — cell, aspect, Self — as Paper 2's architectural primitive, so that downstream work can adopt, extend, compose with, or argue against the framework without ambiguity.

Abstract

Paper 2 of the Coordination Knowledge Substrate (CKS) theory series introduces three architectural levels at which CKS-governed AI Selves are structured: the cell (the atomic unit from Paper 1), the aspect (a coordination arrangement of cells serving a particular purpose), and the Self (the integrated whole holding multiple aspects as facets of one CKS-governed intelligence). The three levels are the structural framework within which Paper 2's other commitments operate. This note formalizes the three-level structure as a standalone foundational commitment with six properties: distinct scope at each level, relational and purpose-defined roles, higher-levels-operate-over-lower-as-content-domain, non-strictly-hierarchical access, structural evolvability under governed reorg, and recursive Paper 1 commitments at every level. The note states what the framework requires architecturally and what it does not, what makes it distinguishable from rigid hierarchical and single-level architectures, and the points where it borrows biological vocabulary while distinguishing itself from biology in architectural substance.

1. Why the three-level structural framework needs to be formalized as standalone

Paper 2's other architectural commitments — the instinct/reasoning separation note B1.01 formalizes, lifecycle as governed primitives, evolution as three mechanisms in productive tension, multi-shaped governance — all operate at and across three structural levels. Lifecycle primitives apply at every level. Evolution mechanisms operate at the levels appropriate to them. The instinct/reasoning separation extends from cell scope to Self scope. Bidirectional evolution distinguishes horizontal change within a level from vertical change across levels. Each commitment presupposes that the structural framework exists and has known properties.

If the framework is treated implicitly — taken for granted in the development of lifecycle and evolution but never separately articulated — three failure modes follow. Implementations conflate the levels and lose the per-level scope distinctions Paper 2's machinery requires. Critics treat the framework as either a rigid hierarchy (which Paper 2 refuses) or as decorative naming (which it equally refuses). Composition with adjacent architectures proceeds without a precise statement of what the levels are and what they require.

This note is the second of approximately twenty Phase B1 foundational notes, following B1.01. It introduces the framework including its key properties; subsequent notes elaborate each level individually and formalize each property as standalone commitment.

2. The architectural commitment, precisely stated

A CKS-governed AI Self is structured at **three architectural levels**, each with distinct scope and purpose, related by content-domain rather than command-and-control, with structural roles that are relational and purpose-defined rather than intrinsic.

Cell. The cell is the atomic unit Paper 1 (Li, April 2026) defends — a CKS artifact with substrates, orchestration rules, and the architectural commitments Paper 1 establishes (substrate-as-coordination-artifact, human-governed authority, conflict preservation, AI-as-substrate-mediator, tool-agnosticism, linear-cost). All six commitments hold at the cell level. The cell handles a specific informational task; it is the level at which the substrate/LLM division at the governance boundary is drawn.

Aspect. An aspect is a coordination arrangement of cells serving a particular purpose. Multiple aspects can coexist within one Self, potentially sharing cells across arrangements. An aspect operates over its constituent cells as content domain — asking pattern questions across the cells the arrangement calls on. Its distinctive scope is purpose-defined coordination across cells.

Self. The Self is the integrated whole that holds multiple aspects as facets of one CKS-governed intelligence. The Self is not a bigger aspect, not a top-of-hierarchy, and not one end of a strict access hierarchy; it contains the multiple coexisting structural perspectives as one unified whole. The aspects are facets it holds, not subordinates it directs.

Relational and purpose-defined roles. The same underlying CKS artifact can participate as a cell in one structural arrangement, as part of an aspect in another, or as a component of different aspects simultaneously, depending on what the structure is for. Roles are role-instances that inhere in the bearer per the arrangement that calls on them; they are not intrinsic properties. A given cell appearing in three aspects is one cell with three role-instances, not three different cells.

Higher-levels-operate-over-lower-as-content-domain. Aspects do not command their constituent cells in a hierarchical sense; they use the cells as content domain for the pattern questions their purpose calls for. The Self holds the aspects as the multiple coexisting perspectives whose collective intelligence is what the Self carries.

Non-strictly-hierarchical access and structural evolvability. The Self can access cells directly when purpose requires, without first traversing aspects. Arrangements themselves are evolvable: cells can be

reassigned across aspects; aspects can split, merge, or dissolve; new aspects can be introduced. Structural arrangements are evolvable artifacts under governed reorg, not fixed schemes baked into the architecture.

Recursive Paper 1 commitments at every level. Paper 1's six architectural commitments hold at the cell level by direct inheritance and at the aspect and Self levels by recursive extension. The substrate/LLM division at the governance boundary holds at every level. Human-governed authority, conflict preservation, tool-agnosticism, AI-as-substrate-mediator, and linear-cost properties extend through the levels by recursion rather than by progressive dilution.

3. What makes the three-level structural framework architecturally distinctive

Two contrasts define what is at stake.

Single-level architectures. The dominant LLM-only architectural shape collapses the system into one level: one model, one system, one population of behavior. Errors, capabilities, coordination, and governance all live at the same level. The three-level framework is non-collapsing by commitment: distinct scope at each level is a design property, not an emergent organization.

Rigid hierarchical orchestration architectures. Many multi-component architectures impose strict hierarchy: an orchestrator commands subordinate tools; a planner directs executors; a top-level agent dispatches sub-agents. Access flows only along the hierarchy. The three-level framework is non-strictly-hierarchical by commitment: the Self can access cells directly when purpose requires, aspects relate to cells as content domain rather than as commanders, and the same cell can participate in multiple aspects without being owned by any one of them.

The architectural distinctness rests on two conjoined properties: relational role flexibility — the same artifact participating in different arrangements as different role-instances — and recursive commitments — Paper 1's commitments holding at every level by inheritance rather than being progressively diluted. Either property alone is recognizable from adjacent architectures; the conjunction is specific to CKS.

4. The biological analog and the points where CKS distinguishes itself

The framework loosely parallels the biological cell–tissue–organism progression, with the parallel functioning as conceptual scaffold readers absorb quickly. The architectural substance distinguishes CKS at three specific points.

Relational rather than intrinsic membership. Biological cells are fixed in their tissue and organ memberships through development; a hepatocyte does not also participate as a cardiomyocyte simultaneously. CKS cells participate in multiple aspects simultaneously through relational role-assignment, with the same artifact carrying multiple role-instances under multiple arrangements concurrently.

Structural evolvability on operational timescales. Biological body plans take millions of years to evolve. CKS structural arrangements evolve through governed reorg on operational timescales, within bounds humans set.

Modular from the start rather than evolved. Biology had to evolve modular cell architectures over deep time. CKS is modular by architectural commitment from the start at every level — a design requirement at instantiation time rather than a property the architecture hopes for or measures after the fact.

The architectural work the framework does is CKS's own. The biological progression supplies vocabulary; what CKS commits to at each level is independently specified by the recursive Paper 1 commitments and the relational role property.

5. Inherited Paper 1 commitments at every level

The recursive-commitments property binds the framework to Paper 1. Four inheritance edges are load-bearing for B1.02 and are formalized in their own right elsewhere in the prior-art chain.

Human-governed authority (Paper 1 §2.1, §3.3; Series A note A1.01). Humans retain the rights to inspect, modify, and override substrate content and orchestration rules at all times. The rights hold at the cell level by direct inheritance, at the aspect level because the coordination arrangement of cells is itself substrate content under human authority, and at the Self level because the composition relationships across aspects are themselves substrate content under human authority.

Substrate-cell boundary (Paper 1 §2.1, §4.1; Series A note A1.02). The two-layer separation between substrate (coordination, governance, conflict-handling) and cell (executable behavior under orchestration rules) holds at the cell level by direct inheritance and extends to aspect and Self levels through their constituent cells; the aspect's coordination logic and the Self's composition logic are themselves substrate content over which the boundary holds.

Composition requirements (Paper 1 §2.1, §3.1, §6.1, §13.3; Series A note A1.13). The five constraints any multi-substrate composition must satisfy under Paper 1 — what counts as a primitive, what crosses substrate boundaries, what is preserved, what governance properties persist, what addressability is maintained — extend across levels. Composition within an aspect, and composition between aspects within a Self, hold at their respective scopes.

Hybrid systems composition (Paper 1 §2.3, §6.1, §6.2; Series A note A1.16). The three composition patterns by which CKS substrates compose with adjacent AI components — input, derived view, separate concern — hold at every level. Each level has substrate handling coordination, governance, and reasoning, and LLM handling instinct, with the division at the governance boundary as Paper 1 specifies.

These edges are stated here as inherited references. Series C notes formalize each inheritance edge from Paper 2 to Paper 1 explicitly.

6. Operational implications

Three operational implications follow from naming the framework as standalone foundational commitment.

Lifecycle and evolution operate at the appropriate level. Birth, mating, and death apply at every level: cells, aspects, and Selves are each born, mate, and die. The three evolution mechanisms operate

at the levels where they are appropriate. Bidirectional evolution distinguishes horizontal change within a level from vertical change across levels. Each operation is named at the level where it operates.

Deployments organize by recognizing and naming levels rather than by inventing structure. Operators recognize which arrangements are aspects and which are Selves, name the role-instances their cells carry, and expose the structure to humans for governance. Conflating levels — discussing what an aspect "does" as if it were a bigger cell, or treating the Self as a top-of-hierarchy commander — produces architectural confusion the framework prevents.

Vertical evolution operates on the structural arrangements themselves. Because the arrangements are evolvable, the framework supports a class of governed change that horizontally-scoped architectures cannot express: cells reassigning across aspects, aspects splitting or merging or dissolving, new aspects being introduced. This is the kind of structural co-adaptation that distinguishes CKS from architectures whose composition relationships are baked at design time.

7. Limits

The framework is bounded in four respects.

Not a rigid hierarchy. The levels are not ordered by command authority. Higher levels operate over lower levels as content domain rather than directing them in a chain-of-command sense, and the Self can access cells directly when purpose requires.

Not permanent. The structural arrangements are not fixed for the lifetime of the Self. The framework supports governed reorg as architectural property.

Not eliminating cell-level work. Cells remain the level at which informational tasks happen. Aspects and Selves coordinate cells and operate over them as content domain; they do not absorb cell-level work into higher abstractions.

Not exhaustive. The framework does not claim that three levels are the only levels possible. Additional levels may emerge or be introduced where purpose requires — sub-cell decomposition for very large cells, super-Self arrangements at organizational scope. The three named levels are the foundational primitives Paper 2 commits to; extensions are describable in the same recursive-commitments and relational-roles vocabulary.

8. Operational test

A system instantiates the CKS three-level structural framework if and only if all of the following hold at all times:

- The system distinguishes cells (atomic CKS artifacts under Paper 1 commitments), aspects (coordination arrangements of cells serving particular purposes), and Selves (integrated wholes holding multiple aspects as facets), with the distinctions named in the substrate.
- Each level's distinctive scope is preserved: cells handle specific informational tasks; aspects coordinate cells for purpose-defined engagement; Selves hold multiple aspects as one whole.

3. Structural roles are relational and purpose-defined: the same underlying CKS artifact can participate as a cell in one arrangement and as part of an aspect in another, simultaneously, depending on what each arrangement is for.
4. Higher levels operate over lower levels as content domain rather than by hierarchical command, and access patterns are not architecturally restricted to strict hierarchy.
5. Paper 1's six architectural commitments hold at every level — by direct inheritance at the cell level and by recursive extension at aspect and Self levels.
6. Structural arrangements are evolvable under governed reorg rather than fixed at design time.

A system that fails any of (1)–(6) may be a useful system, and may be governed in some other sense, but is not CKS-coherent on the three-level structural axis as Paper 2 commits to it.

9. Why naming the three-level framework as standalone matters

Conflating the three levels into a single composite makes Paper 2's other commitments difficult to express precisely. Lifecycle primitives become ambiguous (which level is being born?), evolution mechanisms become unclear (what is changing where), and governance becomes unscoped (who has authority over what). Naming the framework as standalone foundational commitment is what gives downstream commitments a precise place to operate.

Subsequent Phase B1 notes elaborate the framework in two directions. Level-elaboration notes formalize each level on its own terms (cell as atomic Paper-1 unit; aspect as purpose-defined coordination arrangement; Self as integrated whole). Property-elaboration notes formalize each structural property as standalone commitment (relational-and-purpose-defined roles; higher-as-content-domain; cross-level access; recursive Paper 1 commitments). Phase B1 then progresses through dynamic primitives (lifecycle and evolution) and governance primitives. Each subsequent commitment operates over or within the three-level framework B1.02 names here.

Subsequent work that adopts, extends, composes with, or argues against the CKS three-level framework should use it in the sense formalized here. Work that uses the levels differently — as rigid hierarchy, as fixed arrangements, as intrinsic memberships, or as decorative naming — is using a different concept, and the difference should be named.

Source paper

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